BRAC UNIVERSITY

Inspiring Excellence

**CSE340**

Computer Architecture

Research-based Assessment

Section: 04

Semester: Spring_2026

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Submitted Date: 01-05-2026

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# Parallel Inference Engine for Autonomous Disaster Drones

## Introduction

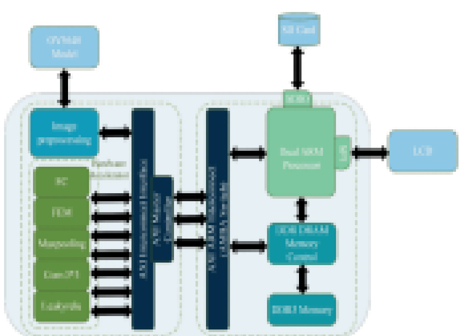
As the name suggests Autonomous Disaster Drones are employed to locate the survivors and map the terrain affected by natural disasters like flooding, earthquakes and building collapses. The search and rescue of the survivors has to be carried out as quickly as possible, and for that to happen, the post-disaster drones need to be fast and smart, which is a challenging task for computer architecture. This report clearly describes a lightweight parallel CNN architecture with a stringent consideration of cost-effectiveness(less delay,power, memory traffic) and reliability.While keeping energy efficiency and fault tolerance this report mainly focuses on incorporating SPMD(Single program,multiple data) dependent parallel processing, sharing CNN weights, and reusing SRAM or cache.

## Background/Related Work

Architectures for disaster response drones must combine fault tolerance, hardware acceleration and energy efficiency. The HeRO architecture fosters an "expect and detect failures" approach in terms of redundant state estimation and supervisory recovery (Santamaria-Navarro et al.), complemented with mission reasoning (Ortiz Barbosa et al.) and group-level fault-tolerance (Khan and Amir). Hardware-wise, FPGA-based accelerators with multi-scale parallelism support efficient, fast CNN inference (Wang et al.), manycore MPPA3-based processors enable SPMD execution with distributed memory (de Dinechin et al.), and near threshold voltage operation enables performance scaling with minimum energy consumption (Nunez-Yanez and Howard).

Algorithmically, efficient real-time inference system is achieved with MobileNet's depthwise separable convolutions and SVD-compressed Long Short-Term Memory (LSTM) on FPGA to obtain deterministic, low-latency control (Priya et al.). This potential is supported by networking and task allocation that enables multi-drone disaster response scenarios (Yanmaz et al.).

## Methodology / System Design



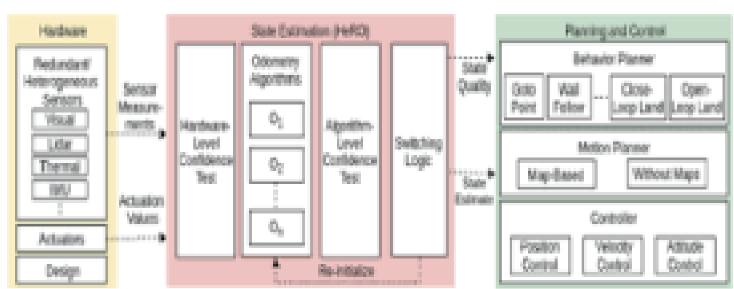




Figure 1: System Architecture of onboard FPGA(Wang et al.) Figure 2: General resiliency Architecture(Santamaria-Navarro et al.)

**Architecture**: 4-core ARM Cortex-A55 (control/pre/post-processing) + 16-core RISC-V SPMD inference array + small NPU for key convolutions.

**Execution Model:** One tile of the feature-map per core with the same CNN kernel (SPMD spatial tiling). **Target Model:** MobileNetV2-SSD Deep Separable Convolutions (300M MACs).320×320 RGB, ≤2W total power, GPS-degraded and high-vibration operation conditions.

**Memory Layout & Tiling Strategy, Fault Tolerance**

| Section | Item/Layer | Configuration/value | Contents/Mec h anism | Access Pattern/Recovery |
|---|---|---|---|---|
| Memory Layout and Tiling | L1 Scratchpad (per core) | 32KB | Input tile (5x5 to 16x16 patches),parti al sums | Streaming double-buffer |
| | Shared L2 SRAM | 512KB | Current layer weights, bias vectors | Broadcast to all cores |
| | LPDDR4 DRAM | 512MB | Full model weights, input frame buffer | Layer-by-layer prefetch |
| Fault Tolerance | Data Integrity | ECC on all SRAM/DRAM(SECDE D) | | Single-bit correct,double-bit detect, and flag |
| | Core Redundancy | 2 of 16 cores are designated as hot spares | | Failed core's tile reassigned in <1 cycle |
| | Algorithm Health | Confidence testing on detection outputs | | Fallback to simplified model if anomaly detected |

**Tiling Scheme:** 320×320 input split into 16 SPMD cores (80×80) with a halo overlap; weights broadcast through shared L2 instead of reading from DRAM multiple times to save power (Wang et al). **Failure Model:** Based on HeRO's layered fault-tolerance (Santamaria-Navarro et al.).

**Graceful Degradation:**

**Normal:** 16 cores → 320×320 at max framerate

**Degraded-1:** 14 cores → same resolution, ~12% latency increase

**Degraded-2:** <12 cores → 160×160, <150 ms latency

**Emergency:** ARM only → blob detection at low resolution, 2 FPS

**Key Metric Estimates**

| Metric | Rescue Mode | Survey Mode | Justification |
|---|---|---|---|
| Latency | 80-120 ms | 40-60 ms | Rescue: full MobileNetV2-SSD<br>Survey: pruned model,lower resolution |
| Power | 1.8 W | 1.2 W | Rescue: all 16 cores active<br>Survey: 8cores,reduced clock |

| | | | |
|---|---|---|---|
| Cache Miss Rate | 6-8% | 4-5% | Double-buffering + weight broadcast keeps the working set in SRAM |
| Throughput | 8-12 FPS | 16-25 FPS | Rescue: prioritizes accuracy<br>Survey: prioritizes coverage |
| Fault Recovery | <5ms (core swap) | <5ms | Hot spare immediate substitution |

Performance results are consistent with 24.7 ms FPGA CNN results (Wang et al.) and near-threshold efficiencies; rescue mode execution is full resolution and triple modular redundancy and majority voting, emulating swarm-type fault tolerance (Khan and Amir).
Survey powering reduces the cores by one-half and resolution by a factor of four, and doubles the flight time in broad-area surveys.

**Discussion**
**Impact Evaluation & Trade-offs:** The design attains up to ~2-4 TOPS/W through SPMD tiling, near-threshold RISC-V cores, and L2 weight broadcast, which is optimal in terms of CNN regularity. Limitations to non-CNN flexibility, scratchpad capacity constraints, throughput cost of hot spares is about 12.5 percent, and halo overhead is about 12 percent (5x5 kernels) are trade-offs. Swarm task allocation (Yanmaz et al.) and confidence-based multiplexing (Santamaria-Navarro et al.) alleviate limitations and promote safe autonomy.
**Sustainability:** The system also generates energy-per-inference that is 100 times more efficient than cloud offloading with 1.210 -.218W, which you can significantly reduce the fleet power consumption. Embodied or operational energy is minimized by mature process nodes and power-gating and long-term software support is possible with the open RISC-V ISA. This is a trade off between short term effectiveness and lifecycle responsibility, overcoming e-waste and silicon-demand challenges at scale.

**Conclusion**

It is demonstrated here that with a hierarchical memory and spatial tiling, CNN inference can be mapped to a lightweight SPMD architecture with a strict <2W budget to the real-time detector (less than 150 ms) and shared weight caching can be used to resolve memory bottlenecks. A resilience approach that takes the HeRO approach guarantees graceful degradation rather than disaster (Santamaria-Navarro et al.). Future directions include dynamic precision scaling (INT8 or INT4), integration of transformers and formal verification of the fault-tolerance controller to be aviation certified.

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